PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The impact of quality improvement initiatives to improve CKD referral patterns: A systematic review protocol
AUTHORS	Ghimire, Anukul; Sultana, Naima; Ye, Feng; Hamonic, Laura; Grill, Allan; Singer, Alexander; Akbari, Ayub; Braam, Branko; Collister, David; Jindal, Kailash; Courtney, Mark; Shah, Nikhil; Ronksley, Paul; Shurraw, Sabin; Brimble, Kenneth Scott; Klarenbach, Scott; Chou, Sophia; Shojai, Soroush; Deved, Vinay; Wong, Andrew; Okpechi, Ikechi; Bello, A

VERSION 1 – REVIEW

REVIEWER	O'Hanlon, Claire RAND Corporation, RAND Health Care
REVIEW RETURNED	11-Sep-2021

GENERAL COMMENTS	The authors present a reasonable systematic review protocol to answer an interesting and important question. My only major comment is about the exclusion of general internal medicine (or similar) as primary care practitioners. The authors may come across studies that include general internal medicine as primary care practitioners among the eligible referral sources. How will you treat studies that mix these groups? Minor comments: Page 1 line 36: Is this your list of databases or a list of possible databases (as implied by "e.g."?). I think you should specify the databases you will use in the protocol. Page 3 line 12: I think you mean "inform" not "form" Page 3 line 14: I don't understand what it means to "make the objectives of improving CKD referral patterns more meaningful" Page 3 line 22: This is the only place where you mention costs. I would delete this irrelevant point. Page 5 line 20: "Undoubtedly" is too strong since this likely varies by care context Page 5 line 31: "are unnecessary" is not necessarily true. There may be edge cases where a referral is necessary for reasons that care guidelines do not account for. Page 5 line 31: Another reason could be that patients insist on it and it's easier to give in than argue with patients about it. Page 5 line 51: I'm not familiar with the area of "QI analysis." Do you mean just QI? Page 6 line 29: What do you mean by "professional or organizational in nature"? Page 8 line 42: Explain your search strategy in plain language here, referencing the PICOTS framework you use above. Only mention the databases you will actually use. Both uses of "i.e." are wrong here (should be e.g.) Page 9 line 33: What is your decision rule for your initial screening?

	Include if either reviewer yes, or consensus? Page 24: Provide a title for table S1

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Comments to the Author:
The authors present a reasonable systematic review protocol to answer an interesting and important question. My only major comment is about the exclusion of general internal medicine (or similar) as primary care practitioners. The authors may come across studies that include general internal medicine as primary care practitioners among the eligible referral sources. How will you treat studies that mix these groups?

Thank you for your positive feedback. Regarding your major comment, we have updated the relevant section of the manuscript (page 7, line 17) making this important point more explicit:

"…we will exclude studies that include general internists and/or pediatricians as primary care providers among eligible referral sources. We chose to exclude such studies given that our focus is to assess the impact of implementing QI on referrals from primary care to nephrology. Including studies with referrals from general internal medicine (GIM) and pediatrics could create heterogeneity among the studies and confound our outcome and conclusions given that GIM and pediatrics, in many places, are considered specialists rather than primary care. It is reassuring that this category of referrals from general medicine and pediatrics are small (<10% of all referrals to nephrology) based on empirical information available to use, and the referral model within the Canadian health system where primary care providers (family physicians and GPs) constitute the main gatekeepers to specialist care."

Minor comments:
Page 1 line 36: Is this your list of databases or a list of possible databases (as implied by "e.g."?). I think you should specify the databases you will use in the protocol.

This has been done. A list of the databases we will search is now updated based on this suggestion. We have removed "e.g.," to ensure clarity.

Page 3 line 12: I think you mean "inform" not "form"

Thank you. This has been done. We have changed the wording to "inform" as corrected.

Page 3 line 14: I don't understand what it means to "make the objectives of improving CKD referral patterns more meaningful"

This part has now been made clearer in the text of the manuscript under summary points as follows: "Our study findings can inform focus groups in the future that will incorporate opinions of patients, policy makers, and scientific researchers to further explore methods in enhancing referral patterns from primary care to nephrologists"

Page 3 line 22: This is the only place where you mention costs. I would delete this irrelevant point.

Thank you. We have deleted this.

Page 5 line 20: "Undoubtedly" is too strong since this likely varies by care context

Thank you. We have deleted this term as previously used in the text.

Page 5 line 31: "are unnecessary" is not necessarily true. There may be edge cases where a referral is necessary for reasons that care guidelines do not account for.
Page 5 line 31: Another reason could be that patients insist on it and it's easier to give in than argue with patients about it.

Thank you. We have deleted "are unnecessary" and kept the statement "… approximately 40% of referrals to nephrology for CKD management are not concordant with guidelines" to reflect adherence to guideline criteria.

Page 5 line 51: I'm not familiar with the area of "QI analysis." Do you mean just QI?

Yes correct, we meant just QI. We have removed "analysis"

Page 6 line 29: What do you mean by "professional or organizational in nature"?

We have updated to clarify both terms on page 6 line 12

Professional: defined as interventions such education for PHC, information provision, or guidelines

Organizational: defined as primary healthcare and specialist provider schemes, general practitioner fundholding schemes, and open access referral schemes.

Page 8 line 42: Explain your search strategy in plain language here, referencing the PICOTS framework you use above. Only mention the databases you will actually use. Both uses of "i.e.," are wrong here (should be e.g.)

Thank you for this comment. We have updated the statement for clarity, It now reads as shown below (page 9 line 5):

"We have developed a search strategy in consultation with our research librarian (LH) (Appendix 1). We will search the following electronic databases — MEDLINE, Embase, Cochrane Library, CINAHL, Web of Science, and PsycINFO—using a combination of controlled vocabulary search terms; the MEDLINE search strategy is shown in Supplementary Table S1."

Page 9 line 33: What is your decision rule for your initial screening? Include if either reviewer yes, or consensus?

We have updated this statement to improve clarity in page 10 line 3 in the manuscript

"For both the first and second stages of screening, studies will be included if there is consensus between the two reviewers. If there is a disagreement, a third reviewer (IO) will resolve such conflicts and make decision on eligibility."

Page 24: Provide a title for table S1
The title "Medical Subject Headings for MEDLINE database search" has been added

VERSION 2 – REVIEW

REVIEWER	O'Hanlon, Claire RAND Corporation, RAND Health Care

REVIEW RETURNED	02-Apr-2022

GENERAL COMMENTS	My concerns have been adequately addressed.